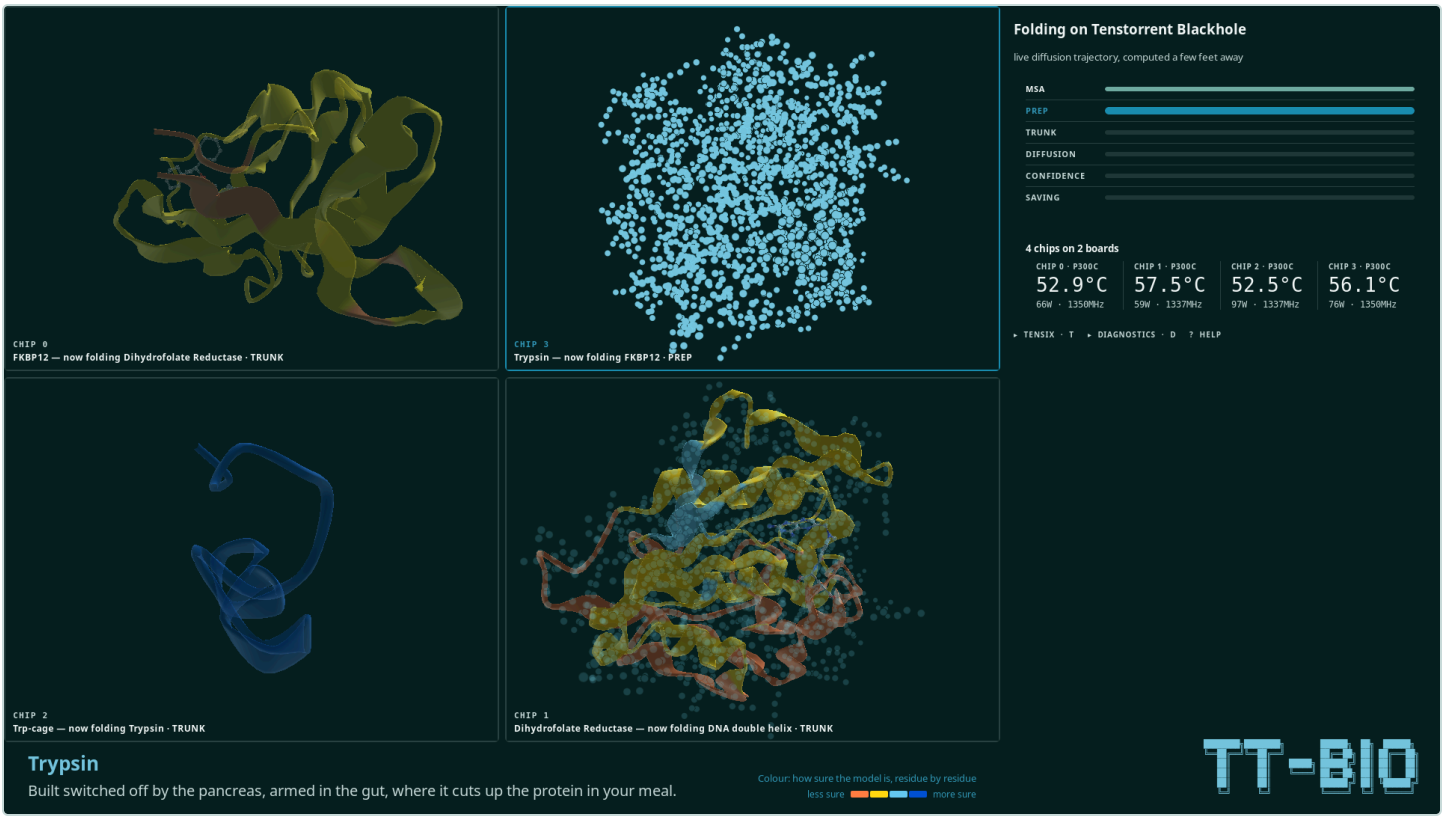
# Watch a protein fold

Live on **Tenstorrent Blackhole**
tt-bio-demo · v0.8.0

A protein is a chain of amino acids that only works once it has folded into a specific three-dimensional shape. Predicting that shape from the sequence alone was an open problem in biology for fifty years. This booth does it in front of you — **on the four Blackhole chips a few feet away**, in seconds, with nothing pre-recorded.




**The real application, on real silicon.** *Four Blackhole chips (two p300c boards), each folding a different molecule at its own point in the pipeline. Structures are drawn as a cartoon — helices as ribbons, sheets as arrows. The loose dots are the model's actual denoising trajectory, streamed one step at a time as it condenses into structure.*

## WHAT YOU ARE WATCHING

The model starts from **pure noise** — a cloud of atoms in random positions — and removes a little of that noise at a time, guided by what it has learned about how real proteins are built. Every dot on screen is a genuine intermediate step, sent from the chip as it happens.

Nothing here is an animation of a finished answer. When a fold has no coordinates yet, the booth holds the **previous real structure**, dimmed and labeled, rather than inventing something to show you.

## COLOR = THE MODEL'S CONFIDENCE

| very high | confident | low | very low |
|---|---|---|---|

Each residue is colored by **pLDDT** — the model scoring its own certainty. Blue regions it is sure about; orange it is guessing. An honest model tells you where it is weak, and this one is shown doing exactly that.

## THE SIX STAGES ON THE PANEL

| | |
|---|---|
| `MSA` | related sequences from other organisms — skipped here, which is why some targets come back orange |
| `PREP` | builds the model's input features |
| `TRUNK` | how every residue relates to every other. Makes **no coordinates** — so a big protein shows nothing new at first, and the booth keeps the previous structure up |
| `DIFFUSION` | the part you watch: noise becomes structure |
| `CONFIDENCE` | the model scores itself — where the colors come from |
| `SAVING` | writes the result out as mmCIF |

## WHAT IT FOLDS

| MOLECULE | SIZE | WARM |
|---|---|---|
| Trp-cage | 20 res | 3.0 s |
| DNA double helix | – | 3.1 s |
| Transfer RNA | – | 3.5 s |
| FKBP12 + binder | 107 res | 4.4 s |
| DHFR + methotrexate | 187 res | 5.3 s |
| Trypsin + benzamidine | 223 res | 5.9 s |
| Albumin the long one | 585 res | 16.1 s |

*Measured on this hardware, warm, with Protenix v1. Three fold a* **bound ligand** *alongside the protein — including DHFR with methotrexate, a chemotherapy drug, shown sitting in the pocket it blocks. Albumin is the long one: at 585 residues it is what a whole, real, drug-carrying protein costs.*

> **Four chips, four proteins — not one protein four times faster.**
>
> A single fold runs on a single chip; that is tt-bio's documented limit, not something this demo works around. Four chips buy four folds at once, and a shorter wait for whatever you pick.

**The science is not ours.** This booth exists to show other people's work running on Tenstorrent hardware. Folding is **tt-bio** (MIT) by Moritz Thüning; the demo around it is Apache-2.0. Structures are the model's own output, written as mmCIF and drawn here — no reference structure is ever shown in its place.

`github.com/tsingletaryTT/tt-bio-demo`

# Operating the booth

For the person **running** it
tt-bio-demo · v0.8.0

## STARTING IT

```
# the normal way — double-click
# "tt-bio Protein Folding Demo"

# or, from a terminal:
/opt/tt-bio-demo/scripts/run-demo.sh
/opt/tt-bio-demo/scripts/run-demo.sh --solo
```

A booth with more than one chip **comes up in the 2×2 grid**; one chip comes up showing a single large protein. `--solo` forces the single view either way, `--quad` forces the grid, and **Q** toggles at runtime. The launcher starts **both** halves — the compute daemon and the UI — and tears the daemon down when it exits.

## STOPPING IT

**Ctrl** + **Q**, or **Ctrl**-**C** in its terminal. One INFO line about a `tt-smi` sample killed by a signal is **normal** on a clean stop.

Signaling by pid does nothing — the launcher's trap cannot run until the UI returns. Signal the whole process group:

```
kill -INT -$(ps -o pgid= -p <pid> | tr -d ' ')
```

## KEYS

| | |
|---|---|
| **?** or **F1** | the help card, from any screen |
| **Q** | quad view — all four chips at once |
| **T** | Tensix activity — live core grids |
| **D** | diagnostics — the live protocol log |
| **Esc** | close the card, or either rail panel |
| **Ctrl**+**G** | easter egg: the Tenstorrent mark, computed on a chip |
| **Ctrl**+**F** | leave / return to fullscreen |
| **Ctrl**+**A** | restart the booth with affinity Q&A enabled (one-way; no-op if already on) |
| **Ctrl**+**Q** | quit |
| any other key or a tap anywhere | wake the booth and show what it folds |

## RUNNING IT UNATTENDED

For a booth that must survive all day, let systemd own the compute daemon so it comes back by itself:

```
systemctl --user enable --now tt-bio-demo
systemctl --user status  tt-bio-demo
systemctl --user restart tt-bio-demo
```

It restarts **on failure only** — if you stop it deliberately it stays stopped. The UI is still launched separately, from the desktop entry.

## THE RIGHT-HAND RAIL

**Pipeline**
One row per stage — msa, prep, trunk, diffusion, confidence, saving. The bright row is running now; diffusion owns most of the bar. Clears itself after 20 s of silence, so a dead daemon cannot leave a bar frozen all day.

**Chips**
Temperature, power and clock per chip, from a `tt-smi` snapshot every two seconds — sampled on the UI's own thread, never through the daemon, so a wedged daemon still leaves the hardware readout live.

**Tensix activity – `T`**
One animated core grid per chip. Only the chip actually running the fold animates the work, and the header names it.

**Diagnostics – `D`**
Every protocol event as it arrives, with a plain-language line for each stage. Closes itself after five minutes of no input.

**Help card – `?`**
Works from any screen without pausing the fold. Closes after a minute.

## WHEN A VISITOR TAPS A PROTEIN

It gets folded **next** — never "now". The tap goes to the head of the queue and lands on the next chip to come free. Nothing already running is canceled, because tearing a fold down mid-device-operation is a known instability source.

Measured: **0.25–5.5 s** with all four chips busy (median ≈ 2 s). And most taps queue nothing at all — with seven targets on four chips the protein is usually already folding, so that cell simply takes focus. The booth acknowledges every tap immediately on a one-line notice.

> **Never run `tt-smi -r` on a shared machine.** A chip missing from telemetry may have been deliberately quarantined by the daemon for temperature. Check `tt-smi -s` first.

> **Do the first fold before doors open — up to about 75 s per target.** Every time on the front of this sheet is measured **warm**. On a machine that has never folded that target, the kernels compile first: the first fold of each shape took **22.8–74.7 s** against 3.0–16.1 s warm. Fold each one you plan to show, once, while nobody is watching.

## QUICK FIXES

| | |
|---|---|
| A cell says **TRUNK** and looks frozen | Working as designed — trunk makes no coordinates. It is holding the last real structure and says so. |
| A chip missing from the chips panel | Check `tt-smi -s`. It may be quarantined for temperature on purpose. |
| Panels frozen, protein still moving | Telemetry is sampled separately from the daemon. Restart the daemon; the UI can stay up. |
| Everything stopped | **Ctrl**+**Q**, then relaunch from the desktop entry. |

## BEFORE DOORS OPEN

- [ ] **Four chips present.** `tt-smi -s` lists four entries across two `board_id`s. Three means one did not come up.
- [ ] **Fold each target once.** This is the one that gets skipped. It warms the kernel cache so no visitor watches a cold compile.
- [ ] **Help card opens.** Press **?**. It is what a visitor reads when you are talking to someone else.
- [ ] **Weights on disk.** The venue is offline — nothing downloads at the booth. `doctor.sh` checks they are present *and* intact; `dpkg -l` only says a package is installed.
- [ ] **Fullscreen.** **Ctrl**+**F**. A multi-chip booth is already in the grid; **Q** toggles.
- [ ] **Leave it alone for a minute.** The attract loop should start folding on its own, unprompted.

**If something looks wrong:** press **D** for the live protocol log — it names the stage and the chip. The UI never displays a stack trace or raw error text by design, so the log is where the detail lives. Install and recovery: **INSTALL.md**. Known gotchas: **docs/followups.md**.

`4 chips · 2× p300c · Ubuntu 24.04`